# Ruolin Li

WiSE Gabilan Assistant Professor
University of Southern California
Civil and Environmental Engineering | Computer Science (Courtesy)

ruolinl@usc.edu (213) 821-4101
ruolinli.me Google Scholar


| 27 | $2.14M | 5 | 20+ |
|---|---|---|---|
| publications & presentations | awarded research support | current SIAS advisees | invited talks |

## Research Summary

I build **game-theoretic, control, and learning foundations for societally-aware interactive autonomy**: autonomous systems that anticipate strategic behavior, coordinate across agents, and improve outcomes for the people and infrastructures they serve. My group develops equilibrium models, optimization and mechanism-design tools, multi-agent learning algorithms, and trustworthy AI agents for mixed-autonomy transportation, mobility markets, robotics, and wildfire evacuation.

At USC, I direct the **Societally-aware Interactive Autonomous Systems (SIAS) Lab**, where we translate these methods into scalable decision systems for safer, more efficient, and more equitable human–AI collectives.

- **Methods:** game theory and equilibrium modeling; control and optimization; multi-agent learning and agentic AI; mechanism design; resilient decision-making.
- **Applications:** mixed-autonomy transportation; mobility platforms and markets; intelligent infrastructure; robotics; emergency response and wildfire evacuation.

## Academic Appointments

**University of Southern California** Aug. 2024–present
WiSE Gabilan Assistant Professor, Civil and Environmental Engineering
Assistant Professor, Computer Science (Courtesy; since Feb. 2026)

**Stanford University** 2023–2024
CARS Postdoctoral Fellow, Aeronautics and Astronautics
Advisor: Marco Pavone

## Education

**University of California, Berkeley** 2018–2023
Ph.D., Mechanical Engineering (Controls)
Dissertation: *Exploring the Potential of Autonomous Vehicles in Mixed Autonomy Systems*
Advisor: Roberto Horowitz

**University of California, Berkeley** 2016–2018
M.S., Mechanical Engineering (Controls)
Advisor: Roberto Horowitz

**Jilin University, China** 2012–2016
B.S., Automotive Engineering, summa cum laude

## Research Grants and Sponsored Projects

| Years | Sponsor and Project | Role | Status | Total Award |
|---|---|---|---|---|
| 2026–30 | **NSF FIRE Award #2623936 – EVAC-AI**<br>Agentic AI and multi-agent decision support for wildfire evacuation. | Lead PI | Awarded | $2M |
| 2027 | **National Center for Sustainable Transportation, U.S. DOT**<br>Strategic interactions in mixed-autonomy fleet operations. | Lead PI | Awarded | $110K |
| 2026 | **Oliveri Sustainable Prosperity Fund**<br>Human–AV interaction. | Lead PI | Awarded | $30K |

## Selected Honors and Awards

**Best Paper Award**, 8th Bridging Transportation Researchers Conference 2026
- *One of two conference-wide awards; sole Western Track recipient.*

Finalist, IEEE Intelligent Transportation Systems Society Best Dissertation Award 2024
**WiSE Gabilan Assistant Professorship**, University of Southern California 2024
CARS Postdoctoral Fellowship, Stanford Center for Automotive Research 2023
**Rising Star in Mechanical Engineering**, Stanford University 2022
**Rising Star in Civil and Environmental Engineering**, MIT 2021
Helene M. Overly Memorial Scholarship, WTS Foundation 2021
Educational Foundation and Aerospace/Test Measurement Division Scholarships, ISA 2021
Graduate Division Block Grant Award, UC Berkeley 2017, 2021–22
Outstanding Graduate, Jilin University 2016
University First Prize Scholarship, Jilin University 2013–15

## Publications

† current or former student advised/co-advised by Ruolin Li.

### Manuscripts and Preprints

**[P5]** Yue Zhao, Mengyuan Li, **Ruolin Li**, Prince Zizhuang Wang, Shuli Jiang, Linsey Pang, Xiongye Xiao, and Xiyang Hu. "CatchBench: When Can an Agent Failure Be Caught?" *arXiv preprint arXiv:2608.22808*, 2026.

**[P4]** Yue Zhao, Xiyang Hu, Zuobin Xiong, Zhangyu Wang, and **Ruolin Li**. "AI4Fire: Evaluating Large Language Models on Wildfire Tasks." *arXiv preprint arXiv:2610.10946*, 2026.

**[P3]** Sheryl Paul, Samuel Williams, Kexin Wang†, **Ruolin Li**, and Jyotirmoy V. Deshmukh. "Topology-Aware Congestion Pricing: Demand Robust Routing using the Forman–Ricci Curvature." *arXiv preprint arXiv:2609.14980*, 2026.

**[P2]** Jiaxin Hou†, Kexin Wang†, **Ruolin Li**, and Jong-shi Pang. "Traffic Equilibrium in Mixed-Autonomy Network with Capped Customer Waiting." *arXiv preprint arXiv:2512.10194*, 2025.

**[P1]** **Ruolin Li**, Philip N. Brown, and Roberto Horowitz. "A Unified Toll Lane Framework for Autonomous and High-Occupancy Vehicles in Interactive Mixed Autonomy." *arXiv preprint arXiv:2403.14011*, 2024.

### Journal Publications

**[J3]** **Ruolin Li**, Negar Mehr, and Roberto Horowitz. "Submodularity of Optimal Sensor Placement Problems for Traffic Networks." *Transportation Research Part B: Methodological*, 171:29–43, 2023.

**[J2]** Stanley W. Smith, Yeojun Kim, Jacopo Guanetti, **Ruolin Li**, Roya Firoozi, Bruce Wootton, Alexander A. Kurzhanskiy, Francesco Borrelli, Roberto Horowitz, and Murat Arcak. "Improving Urban Traffic Throughput with Vehicle Platooning: Theory and Experiments." *IEEE Access*, 8:141208–141223, 2020.

**[J1]** Negar Mehr, **Ruolin Li**, and Roberto Horowitz. "A Game-Theoretic Macroscopic Model of Lane Choices at Traffic Diverges with Applications to Mixed-Autonomy Networks." *Transportation Research Part B: Methodological*, 144:45–59, 2020.

## Refereed Conference Proceedings

**[C10]** Sheryl Paul, Kexin Wang†, **Ruolin Li**, and Jyotirmoy V. Deshmukh. "Agent-Based Evolutionary Dynamics for Mixed Autonomy Weaving Ramps." 6th IFAC Workshop on Cyber-Physical Human Systems (CPHS), 2026.

**[C9]** Haohui He†, Kexin Wang†, and **Ruolin Li**. "To Stay or to Bypass: Unraveling Mainline Vehicles' Aggregate Strategic Decision-Making at Highway Weaving Ramps." IEEE International Conference on Intelligent Transportation Systems (ITSC), pp. 3178–3185, 2025.

**[C8]** Robert Dyro, Matthew Foutter, **Ruolin Li**, Luigi Di Lillo, Edward Schmerling, Xilin Zhou, and Marco Pavone. "Realistic Extreme Behavior Generation for Improved AV Testing." IEEE International Conference on Robotics and Automation (ICRA), pp. 1354–1362, 2025.

**[C7]** **Ruolin Li** and Roberto Horowitz. "Dynamic Routing and Queueing for Human-Driven and Autonomous Vehicles with Responsive Signal Control." American Control Conference (ACC), pp. 144–155, 2022.

**[C6]** **Ruolin Li**, Philip N. Brown, and Roberto Horowitz. "A Highway Toll Lane Framework that Unites Autonomous Vehicles and High-Occupancy Vehicles." IEEE International Conference on Intelligent Transportation Systems (ITSC), pp. 3482–3489, 2021.

**[C5]** **Ruolin Li**, Philip N. Brown, and Roberto Horowitz. "Employing Altruistic Vehicles at On-Ramps to Improve the Social Traffic Conditions." American Control Conference (ACC), pp. 4547–4552, 2021.

**[C4]** **Ruolin Li**, Negar Mehr, and Roberto Horowitz. "The Impact of Autonomous Vehicles' Headway on the Social Delay of Traffic Networks." IEEE Conference on Decision and Control (CDC), pp. 268–273, 2020.

**[C3]** **Ruolin Li**, Jiaxi Liu, and Roberto Horowitz. "A Game-Theoretic Model for Aggregate Lane Choice Behavior of Highway Mainline Vehicles at the Vicinity of On-Ramps." American Control Conference (ACC), pp. 5376–5381, 2020.

**[C2]** **Ruolin Li**, Negar Mehr, and Roberto Horowitz. "An Extended Game-Theoretic Model for Aggregate Lane Choice Behavior of Vehicles at Traffic Diverges with a Bifurcating Lane." IEEE International Conference on Intelligent Transportation Systems (ITSC), pp. 2226–2231, 2019.

**[C1]** Negar Mehr, **Ruolin Li**, and Roberto Horowitz. "A Game-Theoretic Model for Aggregate Bypassing Behavior of Vehicles at Traffic Diverges." IEEE International Conference on Intelligent Transportation Systems (ITSC), pp. 1968–1973, 2018.

## Refereed Conference Abstracts and Papers (Without Proceedings)

**[CA9]** Tyler Davis†, Kexin Wang†, and **Ruolin Li**. "The Regime-Dependent Cost of Equity in Autonomous Mobility-on-Demand Rebalancing." Transportation Research Board (TRB) 106th Annual Meeting, Washington, DC, January 2027.

**[CA8]** Yizhe Guan†, Kexin Wang†, and **Ruolin Li**. "A Wardrop-Equilibrium-Based Model for Multi-Lane Merging Behavior at Highway On-Ramp Areas." Transportation Research Board (TRB) 106th Annual Meeting, Washington, DC, January 2027.

**[CA7]** Yizhe Guan†, Kexin Wang†, and **Ruolin Li**. "Interpretable Cost Modeling for Strategic Driving Decisions via LLM-Guided Symbolic Reasoning." Transportation Research Board (TRB) 106th Annual Meeting, Washington, DC, January 2027.

**[CA6]** Kexin Wang†, Gavin Huang†, Zehao Wang, Jiachen Li, and **Ruolin Li**. "MAPLE: Macro–Micro Aligned Predictive Lane-Change Coordination for Mixed-Autonomy Traffic Efficiency." INFORMS Annual Meeting, 2026; Transportation Research Board (TRB) 106th Annual Meeting, Washington, DC, January 2027. 🏆 **Best Paper Award**, 8th Bridging Transportation Researchers Conference

**[CA5]** Kexin Wang†, Yizhe Guan†, and **Ruolin Li**. "Bounded Mediation: Limiting LLM Execution Authority in Cyber-Physical Systems." NeurIPS AIWILD Workshop, 2026; Transportation Research Board (TRB) 106th Annual Meeting, Washington, DC, January 2027.

**[CA4]** Luyao Niu† and **Ruolin Li**. "Stackelberg Routing in Mixed-Autonomy Networks with Heterogeneous User Preferences." INFORMS Annual Meeting, 2026.

**[CA3]** Kexin Wang†, Haohui He†, and **Ruolin Li**. "When Altruism Meets Autonomy: Managing Weaving Ramp Congestion with Strategic AVs." Transportation Research Board (TRB) 105th Annual Meeting, Washington, DC, January 2026.

**[CA2]** Zhilin Zhang, Y. Sun, Z. Wang, Y. Nie, X. Ma, P. Sun, and **Ruolin Li**. "Large Language Models for Mobility in Transportation Systems: A Survey on Forecasting Tasks." Transportation Research Board (TRB) 104th Annual Meeting, Washington, DC, January 2025.

**[CA1]** Jiwoo Kim, Hyunwoo Ryu, Jongeun Choi, Joohwan Seo, Nikhil Potu Surya Prakash, **Ruolin Li**, and Roberto Horowitz. "Robotic Manipulation Learning with Equivariant Descriptor Fields." RSS Workshop on Symmetries in Robot Learning, 2023.

### Technical Report and Dissertation

**[D1]** **Ruolin Li**. *Exploring the Potential of Autonomous Vehicles in Mixed Autonomy Transportation Systems.* Ph.D. dissertation, UC Berkeley, 2023.

**[T1]** **Ruolin Li**. *SUMI: An Arterial Traffic Performance Measure Tool.* UC Berkeley, 2018.

## Invited Talks

**Societally-Aware Autonomy: Games, Control, and Infrastructure for the Next Generation of Mobility Systems**

– University of California, Los Angeles, Civil and Environmental Engineering Apr. 2026
– University of California, Irvine, Civil and Environmental Engineering Apr. 2026
– University of Southern California, Industrial and Systems Engineering Feb. 2026
– University of Southern California, Computer Science Oct. 2025
– Florida State University, Computer Science Sept. 2025

**When Altruism Meets Autonomy: Managing Weaving-Ramp Congestion with Strategic CAVs**

– Bridging Transportation Researchers Conference Aug. 2025
– International Conference on Continuous Optimization (ICCOPT) July 2025

**Harnessing Autonomous Vehicles for Smarter Traffic Management**

– Workshop on Mixed-Autonomy Transportation, American Control Conference (ACC) July 2025

– University of California, Riverside, Electrical and Computer Engineering May 2025
– University of Southern California, Information Sciences Institute Oct. 2024

**The Potential of Autonomous Vehicles in Future Intelligent Transportation Systems**
– University of Southern California, Civil and Environmental Engineering; University of Southern California, Urban Planning; University of Southern California Institute of Transportation Engineers; Stanford Center for Automotive Research; ETH Zurich Autonomy Talks; Stanford Mechanical Engineering Rising Stars; Google Research; University of Maryland, Baltimore County, Mechanical Engineering; University of Colorado Colorado Springs, Computer Science; University of Michigan, Mechanical Engineering; University of Wisconsin–Madison, Civil and Environmental Engineering; Massachusetts Institute of Technology, Civil and Environmental Engineering Rising Stars 2021–24

## Teaching

**CE 513: Game Theory for Interactive Autonomy**, USC Spring since 2025
Designer and instructor; course evaluation: 4.6/5

**CE 471: Principles of Transportation Engineering**, USC Fall since 2025
Instructor

**ME 132: Dynamic Systems and Feedback**, UC Berkeley Fall 2022
Graduate student instructor

**E 7: Introduction to Computer Programming for Scientists and Engineers**, UC Berkeley Spring 2022
Graduate student instructor

**ECON C110: Game Theory in Social Sciences**, UC Berkeley Fall 2021
Graduate student instructor, course evaluation: 6.13/7 (department mean: 5.73/7)

## Advising and Mentoring

**Current Ph.D. students:**

- Kexin Wang (USC, Civil and Environmental Engineering, Aug. 2025–present; Best Paper Award).
- Jiaxin Hou (USC, Industrial and Systems Engineering, Aug. 2025–present; co-advised with Jong-Shi Pang).
- Luyao Niu (USC, Civil and Environmental Engineering, Aug. 2026–present; Annenberg Fellow, 2026).

**Current master's students:**

- Yizhe Guan (USC, Civil and Environmental Engineering, Jan. 2026–present).

**Current undergraduate researchers:**

- Tyler Davis (USC, Computer Science and Business Administration, Aug. 2025–present).
- Gavin Huang (USC, Computer Science, Sept. 2025–present).

**Former students:**

- Haohui He (USC, Civil and Environmental Engineering, 2024–2025).
- Elinor Zhang (USC, Data Science, 2024–2025).
- Yanqi Luo (USC, Urban Planning, 2024–2025).
- Shreya Nair (USC, Computer Engineering and Computer Science, 2024–2025).
- Roda Hassan (USC, Computer Engineering, 2024–2025).
- Alina Du (USC, Computer Science and Applied Mathematics, Fall 2025).
- Brian Shi (USC, Computer Science, Fall 2025).

- Bethania Dagim (USC, Civil Engineering, Fall 2025).
- Mathias Wulfman (University of California, Berkeley, Mechanical Engineering, 2018–2019).

**Highlights of Student Achievements:**

- Ph.D. student Luyao Niu received the Annenberg Fellowship in August 2026, when she joined SIAS.
- Ph.D. student Kexin Wang received the 2026 ASCE Los Angeles YMF INWED Student Scholarship in her first year.
- Ph.D. student Kexin Wang led the MAPLE paper, which received the BTR 2026 Western Track Best Paper Award.
- Undergraduate researcher Gavin Huang contributed to the award-winning MAPLE project within his first year of research.
- Undergraduate researcher Tyler Davis earned a first-author conference abstract within his first year of research.

## Professional Leadership and Service

**Research Community Leadership:**

– Co-Organizer, Geometric and Spatial Reasoning in Vision-Language Models (GSR-VLM) Workshop at the AAAI Conference on Artificial Intelligence (2027).
– Co-Chair, TrafficFlowBench, 2026 IEEE Big Data Cup Challenge (2026).
– Vice Chair for Social Media, IFAC Technical Committee 9.2 on Social Impact of Automation (2026–present).
– Organizing Committee, 6th IFAC Workshop on Cyber-Physical Human Systems (2026).
– Session Chair, International Conference on Continuous Optimization (ICCOPT) (2025).

**External Reviewer for Journals, Conferences, and Grants:**

- **Journals:**
  – *IEEE Transactions on Automatic Control*
  – *IEEE Robotics and Automation Letters*
  – *IEEE Transactions on Intelligent Transportation Systems*
  – *IEEE Transactions on Control of Network Systems*
  – *IEEE Transactions on Control Systems Technology*
  – *International Journal of Transportation Science and Technology*
  – *Transportation Science*
  – *Transportation Research Part B: Methodological*
  – *Transportation Research Part C: Emerging Technologies*
- **Conferences:**
  – IEEE Conference on Decision and Control (CDC)
  – American Control Conference (ACC)
  – European Control Conference (ECC)
  – IEEE International Conference on Intelligent Transportation Systems (ITSC)
  – AAAI Conference on Artificial Intelligence (AAAI)
  – IEEE International Conference on Robotics and Automation (ICRA)
  – Conference on Neural Information Processing Systems (NeurIPS)
  – IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
  – International Conference on Learning Representations (ICLR)
- **Grants:**
  – National Science Foundation (NSF), Directorate for Engineering, Division of Civil, Mechanical and Manufacturing Innovation (CMMI), Civil Infrastructure Systems (CIS) Program
  – National Science Foundation (NSF) Smart and Connected Communities (S&CC) Program
  – National Science Foundation (NSF), Directorate for Engineering, Division of Civil, Mechanical and Manufacturing Innovation (CMMI), Infrastructure Systems and People (ISP) Program
  – National Science Foundation (NSF) Cyber-Physical Systems (CPS) Program

- National Science Foundation (NSF), Directorate for Engineering, Division of Civil, Mechanical and Manufacturing Innovation (CMMI), Mind, Machine, and Motor Nexus (M3X) Program
- National Science Foundation (NSF) Computational and Data-Enabled Science and Engineering (CDS&E) Program
- National Science Foundation (NSF), Directorate for Technology, Innovation and Partnerships, Translation to Practice (TTP) Program
- Transportation Research Board, National Cooperative Highway Research Program (NCHRP)

**USC Service:**

- **CEE Department Seminar Committee** (Fall 2024–present).
- **CEE Department Ph.D. Achievement Award Committee** (Fall 2025).
- **Screening, Qualifying, Thesis Proposal, and Dissertation Committees (not chair):**
  - Milad Pooladsanj (USC CEE, 2024; Chair: Ketan Savla)
  - Fernando Valladares Monteiro (USC EE, 2025; Chairs: Ketan Savla and Petros Ioannou)
  - Yuehan Qin (USC CS, 2025; Chair: Yue Zhao)
  - Jiageng Mao (USC CS, 2026; Chair: Yue Wang)
  - Haoyan Xu (USC EE, 2026; Chair: Mengyuan Li)
  - Jiawei Yang (USC CS, 2026; Chair: Yue Wang)
  - Curtis Bechtel (USC CS, 2026; Chair: Shaddin Dughmi)
  - Junjie Ye (USC CS, 2026; Chair: Yue Wang)
  - Li Li (USC CS, 2026; Chair: Yue Zhao)
  - Gary Rostomyan (USC EE, 2026; Chair: Petros Ioannou)
  - Wen Ye (USC CS, 2026; Chair: Xuezhe Ma)
  - Chen Chu (USC CS, 2026; Chair: Cyrus Shahabi)
  - Shahriar Tanvir Alam (USC ISE, 2026; Chair: Maged M. Dessouky)
  - Wei Yang (USC CS, 2026; Chair: Yue Zhao)
  - Tiankai Yang (USC CS, 2026; Chair: Yue Zhao)
  - Ruohai Ge (USC CS, 2026; Chair: Gaurav Sukhatme)
  - Linxin Song (USC CS, 2026; Chair: Yue Wang)